Confusion Matrix –SVM

```
[[82  3]
 [26 23]]
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	precision	recall	f1-score	support
0	0.76	0.96	0.85	85
1	0.88	0.47	0.61	49
accuracy			0.78	134
macro avg	0.82	0.72	0.73	134
weighted avg	0.81	0.78	0.76	134

1. How is Precision calculated for Class 1?

Solution:

Formula:

Precision=TP/(TP+FP)

TP = 23

FP=3

23/26=0.88

2. How is Recall calculated for Class 1?

Solution:

Formula:

Recall=TP/(TP+FN)

TP=23

FN=26

Recall = 0.47

3. How is F1-measure calculated?

Solution:

Formula:

F1=2·(Precision*recall)/(Recall+Precision)

=2*(0.88*0.47)/(0.88+0.47)

=0.61

4. Can you derive Accuracy from this report?

Solution:

Formula:

Accuracy=(TP+TN)/Total

TP+TN=82+23=105

Total=134

Accuracy= 0.78

5. What is Specificity?

Solution:

Formula:

Specificity=TN/(TN+FP)

=82 / (82 + 3) ≈ 0.96

6. Can you reconstruct the Confusion Matrix (approx)?

Solution:

From support:

- Class 0 → 82
- Class 1 → 23

From recall:

- Class 1 recall = 0.72 → TP ≈23 , FN ≈ 26
- Class 0 recall = 0.78 → TN ≈ 82, FP ≈ 3

	Pred 0	**Pred 1**
Actual 0	82	3

	Pred 0	**Pred 1**
Actual 1	26	23

7. How is Macro Average calculated-precision?

Solution:

Formula:

Macro Average= (P(1)+P(0))/2

=(0.76+0.88)/2

Macro Average = 0.82

8. How is Weighted Average-Precision calculated?

Solution:

Formula:

WA = P(0) *Support (0) + P(1)* Support(1)/ Total

= 0.76 × 85 + 0.88 × 49) / 134 ≈ 0.87

9. How do precision and recall relate mathematically?

Solution:

• Precision depends on **false positives**

• Recall depends on **false negatives**

10. Which class is performing poorly? Why?

Solution:

Class **1** is performing poorly because:

- Recall = **0.47** → model detects less than half of actual class 1 cases
- This indicates a **high number of false negatives**